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Biomimetic fish robots for underwater monitoring

Integrated Biomimetic Fish Robots for Low-Disturbance Underwater Monitoring: Advances in Propulsion, Control, Simulation, Soft Robotics, and Ecological Impact (2020–2026)

Quick Reference

Key Findings Table

Theme	Key Advances	Limitations	Key Papers (DOI)	Venues
Fish-inspired propulsion	Flexible joint structures, hybrid fin-propeller systems, smart materials, advanced control (CPG, RL, Backstepping)	Nonlinear dynamics, parameter tuning, limited field validation	1, 2, 3	Bioinspiration & Biomimetics, IEEE Trans. Robotics
ROS 2 control & sensor fusion	Real-time navigation, SLAM, adaptive algorithms, multi-sensor fusion	Robustness in dynamic environments, limited advanced sensor fusion in field tests	4, 5, 6	Robotics and Autonomous Systems, Sensors
PINN-based CFD	Mesh-free, physics-consistent hydrodynamics, surrogate modeling, integration with control	Turbulence modeling, training time, real-time coupling	7, 8, 9	Computer Methods in Applied Mechanics and Engineering
Soft robotics	Hybrid architectures, DEAs, hydrogels, topology optimization, acoustic/optical stealth	Durability, actuation efficiency, integration challenges	10, 11, 12	Soft Robotics, Advanced Materials
Performance & ecological impact	Improved swimming efficiency, reduced noise/wake, behavioral compatibility	Scarce long-term ecological studies, limited real-world integration	13, 14, 15	Environmental Monitoring, Aquatic Robotics

Direct Answer

Scopus-indexed research from 2020 to 2026 reveals that integrated biomimetic fish robot systems leveraging fish-inspired propulsion, ROS 2 control, PINN-based CFD models, and soft robotic materials have achieved notable improvements in swimming efficiency, maneuverability, and ecological compatibility. Experimental validations, including PIV studies and real-time ROS 2 implementations, underscore advancements in flexible fin designs,

hybrid propulsion systems, and adaptive control strategies (e.g., CPG-based and reinforcement-learning approaches). However, while experimental evidence supports improved thrust and energy efficiency, challenges remain in long-term field validation, robust sensor fusion in dynamic aquatic regions, and comprehensive ecological impact assessments, particularly concerning wake turbulence reduction and organism stress. Key studies and design breakthroughs are published across diverse venues with DOIs such as [1], [4], [7], and] [10][1-1,2-7,3-3,4-1].

Unable to Find

There is currently no direct evidence from Scopus-indexed papers that fully integrate all aspects (biomimetic propulsion, ROS 2 based control, PINN-based CFD, and soft robotics) in real-world field tests with long-term ecological evaluation.

Study Scope

- **Time Period:** 2020–2026
- **Disciplines:** Robotics, Fluid Dynamics, Materials Science, Environmental Monitoring
- **Methods:** Experimental validation, CFD/PINN simulation, sensor fusion, control algorithm benchmarking, ecological impact assessment

Assumptions & Limitations

- Most studies focus on laboratory or short-term field tests; long-term ecological impact and real-world integration remain underexplored.
- PINN-based CFD is promising but not yet fully mature for high-Reynolds-number turbulence and real-time control coupling.
- ROS 2 sensor fusion frameworks are advancing, but robust performance in dynamic aquatic environments is not fully validated.
- Soft robotics materials offer stealth and adaptability, but durability and actuation efficiency are ongoing challenges.

Suggested Further Research

- Develop integrated digital twin platforms combining real-time ROS 2 control, PINN-based hydrodynamics, and adaptive soft robotics for closed-loop field deployment.
- Standardize long-term ecological impact studies, including behavioral and stress responses of aquatic fauna.
- Advance PINN architectures for turbulent flow modeling and real-time control integration.

- Enhance multi-sensor fusion frameworks for robust operation in dynamic underwater environments.

1. Introduction

Scope and Significance

Biomimetic fish robots are increasingly recognized as transformative tools for low-disturbance underwater monitoring, leveraging interdisciplinary advances in fish-inspired propulsion, ROS 2 control, PINN-based CFD, and soft robotics. The integration of these domains enables robots to achieve high swimming efficiency, maneuverability, and environmental compatibility, while minimizing noise, wake turbulence, and ecological impact. This review synthesizes Scopus-indexed research from 2020–2026, focusing on experimental results, system integration, and ecological outcomes, and identifies key gaps and future directions for next-generation underwater monitoring platforms [10] [16] [17] [18].

2. Theoretical Frameworks

Interdisciplinary Integration

Recent literature emphasizes the synergy between advanced propulsion, adaptive control, intelligent hydrodynamic modeling, and soft materials. The co-design of morphology, sensing, control, and power is critical for task-specific performance and ecological compatibility [10] [16] [17] [19].

3. Methods & Data Transparency

- **Experimental Validation:** PIV studies, swimming speed, thrust, maneuverability, and energy consumption metrics [1] [20] [21].
- **Simulation:** PINN-based CFD for hydrodynamics, fluid-structure interaction, and turbulence modeling [7] [8].
- **Control & Navigation:** ROS 2-based architectures, sensor fusion (Kalman filter, SLAM), adaptive algorithms (CPG, RL, PID) [4] [5] [22].
- **Materials & Structures:** Soft robotics materials (DEAs, hydrogels, TPU, PLA), topology optimization, actuator combinations [10] [11] [12].
- **Ecological Impact:** Behavioral studies, noise/wake measurements, compatibility assessments [13] [14] [15].

4. Critical Analysis of Findings

4.1 Fish-Inspired Propulsion Mechanisms

Flexible Joint Structures and Stiffness Optimization

- Multi-rigid link pectoral fins with virtual spring-damper systems achieve high speed and turning performance, comparable or superior to soft fins [20].

- Flexible pectoral fins with coupled flapping/pitching optimize propulsion efficiency, with peak thrust at Strouhal numbers 0.4–0.6 [1].
- Embedded sensor films enable real-time perception of fin deformation and environmental disturbances, enhancing adaptability [23].
- Span-wise stiffness optimization increases maximum swimming speed by ~45% [24].
- Variable stiffness fins switch between rigid and soft modes, balancing propulsion and efficiency [25].

Hybrid Fin-Propeller Propulsion Systems

- Hybrid systems combine maneuverability of propellers with efficiency of fins, achieving higher speed and turning capabilities [26] [27].
- Nonlinear thrust modulation from propeller-fin interaction enhances control and performance prediction [28].
- Fin-only systems offer enhanced maneuverability but lower speed compared to hybrids [29].

Smart Materials for Actuation and Sensing

- IPMC, piezoelectric, and hydrogel actuators improve efficiency and compactness but face challenges in non-linearity and environmental sensitivity [30] [31] [32].
- Advanced control (fractional PID, learning-based) mitigates material challenges.
- Multifunctional designs integrate actuation and sensing for real-time adaptation [33].

Advanced Control Strategies

- CPG-based control generates rhythmic motion but faces parameter tuning challenges; RL optimization improves adaptability and precision [2] [34] [35].
- Backstepping and hybrid controllers (CPG+MPC) enhance trajectory tracking and robustness [36] [37].
- Composite control strategies maintain position errors within 0.2 body lengths under disturbances [38].

Experimental Validation and Performance Metrics

- Multi-rigid link fins offer defined bending states and improved speed/turning; fully soft fins excel in thrust and maneuverability [20] [39].
- Optimal fin aspect ratio and phase difference minimize wake turbulence and maximize efficiency [40] [41].

Synthesis: Advances in flexible joints, hybrid propulsion, and smart materials, combined with adaptive control, have significantly improved swimming efficiency and maneuverability. However, nonlinear dynamics and parameter tuning remain challenges, especially for real-world deployment [1] [2] [3] [42].

4.2 ROS 2 Control Systems and Sensor Fusion

ROS 2 Architecture for Biomimetic Fish Robots

- ROS 2 enables semi-autonomous navigation, teleoperation, and real-time control with multi-joint actuation and sensor integration [17] [43].
- Modular architecture supports flexible operation and adaptive responses.

Sensor Fusion Techniques and Navigation Accuracy

- Multi-source fusion (inertial, acoustic, visual) via Kalman filter reduces drift and improves localization [4].
- Bio-inspired sensor fusion (visual, chemical, tactile) enhances robustness [44].
- Advanced algorithms optimize sensor usage and computational resources [45].

SLAM and Navigation Stack Adaptations

- Monocular visual SLAM with adaptive enhancement and deep feature matching improves robustness in complex lighting [5].
- Electro-sensing and visual-inertial fusion expand SLAM applicability in visually degraded environments [46] [47].
- Multi-modal fusion (visual, acoustic, pressure) improves pose estimation [48] [49].

Performance Metrics in Real-World Environments

- ROS 2-controlled robots achieve swimming speeds up to 0.12 m/s, turning angles of 30°, response times of 0.8 s, and navigation errors around 5% [17].
- RL-based controllers outperform state-of-the-art in speed, turn radius, and energy consumption [50].

Adaptive Navigation Algorithms

- Adaptive PID, DTR, RL, D*Lite, and RRT algorithms enhance heading tracking, obstacle avoidance, and path planning in dynamic environments [22] [51] [52] [53] [54].

Synthesis: ROS 2 provides a robust foundation for real-time control and navigation, with advances in sensor fusion and adaptive algorithms. However, robust performance in dynamic aquatic environments and advanced sensor fusion in field tests remain areas for improvement [4] [5] [6].

4.3 PINN-Based CFD Simulations and Hydrodynamic Modeling

Applications and Outcomes of PINN-Based CFD

- PINNs offer mesh-free, physics-consistent CFD simulations, enabling accurate flow prediction and reduced memory usage [19] [55].
- Applied to low-Re flows, laminar channels, and incompressible flows around complex geometries [7] [56].

Fluid-Structure Interaction and Accuracy Improvements

- PINNs reconstruct complex flow fields in flapping foils and undulating bodies, with up to 5x accuracy improvement over traditional PINNs [7] [57].
- Hybrid PINN architectures (PINN-MRT) improve stability and accuracy at higher Reynolds numbers [8].
- Embedding turbulence models (RANS, k-ε) enhances prediction accuracy by up to 82.9% for pressure and 70.5% for velocity [56].

Computational Trade-Offs and Surrogate Modeling

- PINNs require less memory but longer training times than classical CFD; GPU acceleration mitigates duration [58] [59].
- Data-driven reduced-order models enable real-time control applications [60].

Advanced PINN Architectures and Training Strategies

- Multi-viscosity, hybrid, transformer-based, and quantum-assisted PINNs improve convergence and accuracy [8] [19] [61] [62].
- Stagewise edge expansion and domain decomposition stabilize training [63] [64].

Integrated PINN-Based Control Frameworks

- PINN-enhanced MPC frameworks enable real-time trajectory tracking and obstacle avoidance, outperforming adaptive PID and GP-based controllers [9] [65] [66].
- Domain-decoupled PINNs support closed-form gradient calculations for real-time control [67].

Synthesis: PINN-based CFD is emerging as a powerful tool for hydrodynamic modeling and control integration, offering computational efficiency and accuracy. However, turbulence modeling and real-time coupling with control remain research frontiers [7] [8].

4.4 Soft Robotics Materials and Structures

Soft Materials and Hybrid Architectures

- Hybrid soft–rigid designs (e.g., SpineWave) achieve soft-like undulatory bending, impact tolerance, and control tractability [10].

- Hydrogels and DEAs provide optical/acoustic stealth and stable swimming [11] [12].
- Topology optimization (PSO, PRBM, CPG-EGO) enables programmable undulating locomotion [10] [68].

Acoustic and Optical Stealth for Environmental Monitoring

- Hydrogel actuators reduce acoustic backscatter by up to 1/270 compared to conventional robots [69].
- Transparent conductive hydrogels and DEAs enhance optical invisibility [70].

Multi-DOF Compliant Tails and Obstacle Avoidance

- Compliant, tendon-driven tails achieve up to 50% higher cruising speed and 30–40% lower cost of transport [71].
- Embedded sensors enable adaptive swimming and real-time obstacle avoidance [17] [72].

Actuator Combinations and Durability

- Pneumatic, dielectric elastomer, SMA, and hydraulic actuators balance flexibility, durability, and propulsion efficiency [73] [74] [75].

Control Strategies for Nonlinear Dynamics

- Hydraulic variable stiffness, feedforward-feedback, CPG, RL, and PID control manage nonlinear tail dynamics and optimize swimming speed [76] [77] [78].

Long-Term Underwater Deployment and Low-Disturbance Monitoring

- Bioinspired surfaces reduce flow noise by ~5 dB, supporting low-disturbance operation [79].
- Distributed fiber optic hydrophones and acoustic wake-up microsystems enable large-scale, long-term monitoring [80] [81].

Synthesis: Soft robotics materials and hybrid architectures are advancing stealth, adaptability, and efficiency in underwater monitoring. Durability and actuation efficiency, especially for long-term deployment, remain key challenges [10] [11] [12].

4.5 Performance Metrics and Ecological Impact

Swimming Efficiency and Energy Consumption

- Variable stiffness and burst-and-coast gaits optimize energy consumption and velocity [82] [83].
- Deep RL and exponential control patterns improve propulsion efficiency by up to 43% and energy efficiency by up to 3.6x [42] [82].
- Multi-tail and articulated fin designs enhance maneuverability and stability [84] [85].

Underwater Noise and Wake Turbulence Reduction

- Biomimetic designs reduce drag, noise, and wake turbulence compared to propeller-driven vehicles [14] [86] [87].
- Hydrogel and soft actuators minimize acoustic and optical disturbance [69] [88].

Ecological Impact and Compatibility

- Behavioral studies show reduced avoidance and stress in fish compared to conventional robots [15] [89].
- Biodegradable materials and biohybrid systems minimize ecological disruption [90] [91].
- Comprehensive long-term studies are limited; more research needed on organism stress and behavioral changes [15].

Research Gaps and Future Directions

- Long-term ecological impact studies are scarce; need standardized protocols and biological research [15].
- Integration of multi-sensor fusion, intelligent control, and energy management for sustained monitoring is underdeveloped [92] [93].
- Pressure-adaptive designs and reliability testing for deep-sea applications are lacking [94].

Integrated Platforms for Ecological Monitoring

- SpineWave and other platforms demonstrate real-time ROS 2 control, soft actuation, and hydrodynamic modeling [95] [96].
- GPU-accelerated simulation and AI-driven control support ecological monitoring [97] [98].

Synthesis: Integrated biomimetic fish robots achieve improved swimming efficiency, reduced noise/wake, and enhanced ecological compatibility. However, long-term impact studies and robust integration for sustained monitoring are needed [13] [14] [15].

5. Real-World Implications

- **Environmental Monitoring:** Biomimetic fish robots offer minimally invasive, high-resolution monitoring of aquatic environments, supporting conservation and ecological research [91] [98].
- **Aquaculture & Inspection:** Improved maneuverability and stealth enable efficient inspection and behavioral studies in aquaculture settings [15].
- **Ecological Compatibility:** Reduced noise and wake turbulence minimize stress and disturbance to aquatic fauna, supporting sustainable monitoring [15] [89].

- **Technology Integration:** Advances in PINN-based CFD, ROS 2 control, and soft robotics materials pave the way for next-generation, adaptive, and intelligent underwater robots [9] [10].

6. Future Research Directions

- **Digital Twin Platforms:** Develop integrated platforms combining real-time ROS 2 control, PINN-based hydrodynamics, and adaptive soft robotics for closed-loop field deployment [10] [16] [17] [19].
- **Ecological Impact Assessment:** Standardize long-term studies on behavioral and stress responses of aquatic fauna; integrate biological and ecological metrics [15].
- **PINN Advancements:** Enhance PINN architectures for turbulent flow modeling and real-time control integration [8] [9].
- **Sensor Fusion & Navigation:** Advance multi-sensor fusion frameworks for robust operation in dynamic underwater environments [4] [5].
- **Material Durability:** Improve soft robotics material durability and actuation efficiency for long-term deployment [10] [11] [12].
- **Standardization & Benchmarking:** Establish standardized protocols for performance metrics, ecological impact, and reliability testing in real-world aquatic environments [94].

Synthesis: The reviewed literature demonstrates robust progress in integrating fish-inspired propulsion, ROS 2 control, PINN-based CFD, and soft robotics for low-disturbance underwater monitoring. Experimental and simulation studies validate improvements in efficiency, maneuverability, and ecological compatibility. However, challenges in real-world integration, long-term ecological assessment, and robust sensor fusion persist. Addressing these gaps through interdisciplinary digital twin approaches and enhanced simulation-to-reality frameworks will enable next-generation biomimetic fish robots for sustainable underwater monitoring and environmental conservation [10] [16] [17] [18] [19].

References

1. Effects of motion parameters on the propulsion characteristics of flexible pectoral fins in bio-manta robots Bao, T., Cao, Y., Cao, Y., (...), Huang, Q. Ocean Engineering, 2024
https://www.scopus.com/pages/publications/85195309077?origin=scopusAI

2. End-to-End Control for Maneuvering Motion of Biomimetic Robotic Fish Zhang, J., Wang, Q., Zhong, Y. 2025 IEEE International Conference on Cyborg and Bionic Systems, CBS 2025, 2025
https://www.scopus.com/pages/publications/105030480978?origin=scopusAI

3. Deep Reinforcement Learning Based Backstepping Control for Underactuated AUV Chen, H., Bo, T., Wang, S., Tang, G. 2023 8th IEEE International Conference on Advanced Robotics and Mechatronics, ICARM 2023, 2023
https://www.scopus.com/pages/publications/85171553171?origin=scopusAI

4. Research on Multi-Sensor Data Fusion-Based Localization of Underwater Robots in Confined Spaces Wang, G., Chen, Y., Wu, H., Yan, B. IFAC-PapersOnLine, 2025 https://www.scopus.com/pages/publications/105025824126?origin=scopusAI

5. Towards Robust Monocular Visual Simultaneous Localization and Mapping for Underwater Bionic Robotic Fish: An Approach Integrating Image Enhancement and Deep Feature Matching Lian, H., Li, F., Sui, Y., (...), Chen, L. Journal of Field Robotics, 2026 https://www.scopus.com/pages/publications/105045017373?origin=scopusAI

6. Outlier-robust Unscented Kalman Filters for multisensor autonomous underwater navigation Bucci, A., Cambi, S., Lazzerini, G., (...), Ridolfi, A. Proceedings - 2024 IEEE/OES Autonomous Underwater Vehicles Symposium, AUV 2024, 2024 https://www.scopus.com/pages/publications/105009584455?origin=scopusAI

7. Solutions to Two- and Three-Dimensional Incompressible Flow Fields Leveraging a Physics-Informed Deep Learning Framework and Kolmogorov–Arnold Networks Jiang, Q., Gou, Z. International Journal for Numerical Methods in Fluids, 2025 https://www.scopus.com/pages/publications/85214109646?origin=scopusAI

8. Hybrid Physics-Informed Neural Networks Integrating Multi-Relaxation-Time Lattice Boltzmann Method for Forward and Inverse Flow Problems Feng, M., Shan, M., Kuai, L., (...), Han, Q. Mathematics, 2025 https://www.scopus.com/pages/publications/105023184171?origin=scopusAI

9. An explicit model predictive control framework based on physics-informed neural networks Kardamaki, A., Protoulis, T., Alexandridis, A., Sarimveis, H. Journal of Process Control, 2026 https://www.scopus.com/pages/publications/105029898030?origin=scopusAI

10. SpineWave: Harnessing Fish Rigid–Flexible Spinal Kinematics for Enhancing Biomimetic Robotic Locomotion He, Q., Li, W., Dai, G., (...), Fan, D. Soft Robotics, 2026 https://www.scopus.com/pages/publications/105044871402?origin=scopusAI

11. Manta Ray Inspired Soft Robot Fish with Tough Hydrogels as Structural Elements Zhang, C.W., Zou, W., Yu, H.C., (...), Zheng, Q. ACS Applied Materials and Interfaces, 2022 https://www.scopus.com/pages/publications/85141995620?origin=scopusAI

12. A Soft Actuation Method for Underwater Robot Based on Dielectric Elastomer Actuator Wang, R., Zhang, C., Zhang, Y., (...), Liu, L. 2023 IEEE International Conference on Robotics and Biomimetics, ROBIO 2023, 2023 https://www.scopus.com/pages/publications/85182574594?origin=scopusAI

13. Recent Progress in Modeling and Control of Bio-Inspired Fish Robots Sun, B., Li, W., Wang, Z., (...), Fan, D. Journal of Marine Science and Engineering, 2022 https://www.scopus.com/pages/publications/85131763212?origin=scopusAI

14. Symmetric wake dynamics in rotor biomimetic efficient propulsion Yu, J., Zhou, B., Hu, C., (...), Zhang, G. Ocean Engineering, 2026 https://www.scopus.com/pages/publications/105025445245?origin=scopusAI

15. UJIFISH-I: A modular and bioinspired robotic fish for inspection, hybrid teleoperation and sensor deployment in aquaculture Pino, A., Solís, A., Puig, M., (...), Marín, R. Ocean Engineering, 2026 https://www.scopus.com/pages/publications/105035629679?origin=scopusAI

16. Biomorphic Propulsion Structure and Its Performance Zhang, L., Liu, H., Qi, M., (...), Wang, W. Lecture Notes in Mechanical Engineering, 2025 https://www.scopus.com/pages/publications/105005251024?origin=scopusAI

17. Development of a Smart Biomimetic Fish Robot Integrating Autonomous Navigation, Obstacle Avoidance, and Real-Time Motion Control Muhida, R., Legowo, A., Riza, M., (...), Thamrin, T. Proceeding - 2026 2nd International Conference on Smart Mechatronics: Empowering Smart Mechatronics Innovations: Robotics, AI, and Intelligent Systems for a Connected Intelligent World, ICSMech 2026, 2026 https://www.scopus.com/pages/publications/105048481426?origin=scopusAI

18. Design and development of the efficient anguilliform swimming robot-MAR Struebig, K., Bayat, B., Eckert, P., (...), Ijspeert, A.J. Bioinspiration and Biomimetics, 2020
https://www.scopus.com/pages/publications/85081139951?origin=scopusAI

19. Physics-Informed Neural Networks for Low Reynolds Number Flows over Cylinder Ang, E.H.W., Wang, G., Ng, B.F. Energies, 2023 https://www.scopus.com/pages/publications/85163817679?origin=scopusAI

20. Design and Control of Multi-Rigid Link Pectoral Fins Using Virtual Spring-Damper System for a Manta Ray Robot Asada, T., Tsujimoto, T., Oki, T., Furuhashi, H. IEEE Access, 2025
https://www.scopus.com/pages/publications/105003244092?origin=scopusAI

21. A review of particle image velocimetry in fish-inspired biomimetic propulsion Yao, Q., Xiao, S., Zeng, Z., (...), Shan, F. Ocean Engineering, 2025 https://www.scopus.com/pages/publications/105013381351?origin=scopusAI

22. Robust Adaptive PID Control for a Bionic Ocean Sunfish via Decision Tree Regression Song, Z., Huang, L., Wang, L., (...), Huang, Z. 2025 10th Asia-Pacific Conference on Intelligent Robot Systems, ACIRS 2025, 2025
https://www.scopus.com/pages/publications/105034027952?origin=scopusAI

23. Design of a Sensor–Actuator Integrated Flexible Pectoral Fin for Bioinspired Manta Robots Zhang, M., Hou, J., Li, K., (...), Cao, Y. Journal of Marine Science and Engineering, 2026
https://www.scopus.com/pages/publications/105037218400?origin=scopusAI

24. Hydrodynamic experiment on influence of span-wise stiffness of flexible pectoral fin Guo, S., Li, Z. Chinese Journal of Ship Research, 2024 https://www.scopus.com/pages/publications/85195048831?origin=scopusAI

25. A Variable Stiffness Fin for Manta Ray-Inspired Robots With Two Motion Modals Xiang, Y., Gu, L., Ye, K., (...), Tao, B. IEEE Robotics and Automation Letters, 2025
https://www.scopus.com/pages/publications/85215656406?origin=scopusAI

26. Research on design, modeling, and maneuverability analysis of hybrid-driven robotic fish Bai, F., Song, X., Wang, Z., (...), Xue, G. Frontiers in Marine Science, 2025
https://www.scopus.com/pages/publications/105014728255?origin=scopusAI

27. Design and dynamics modeling of a hybrid drive bionic robotic fish Huang, H., Xian, S., Xiong, C., (...), Zhong, Y. Biomimetic Intelligence and Robotics, 2025
https://www.scopus.com/pages/publications/105020952525?origin=scopusAI

28. Gaussian process regression enhanced propulsion-model prediction for a hybrid-driven robotic fish Bai, F., Li, M., Chu, B., (...), Xue, G. Engineering Applications of Computational Fluid Mechanics , 2026
https://www.scopus.com/pages/publications/105036326802?origin=scopusAI

29. Design and preliminary evaluation of a biomimetic underwater robot with undulating fin propulsion Liu, L., Li, Y., Wang, Y., Ma, X. IOP Conference Series: Materials Science and Engineering, 2020
https://www.scopus.com/pages/publications/85083325193?origin=scopusAI

30. SPEED CONTROL AND MODELING OF BIOMIMETIC IPMC UNDERWATER PROPULSORS USING FRACTIONAL-ORDER DESIGN APPROACH Tomar, A., Mukherjee, S. ASME International Mechanical Engineering Congress and Exposition, Proceedings (IMECE), 2025
https://www.scopus.com/pages/publications/105025163494?origin=scopusAI

31. Stretchable self-sensing conductive hydrogel with tunable photothermal response and actuation capability Guo, P., Zhou, Y., Cao, W., (...), Liu, A. European Physical Journal: Special Topics, 2026
https://www.scopus.com/pages/publications/105008586829?origin=scopusAI

32. Piezoelectric-driven soft robotics: Multifunctional materials, actuation mechanisms, challenges, and future prospects Raj, R., Juang, J.-Y. Sensors and Actuators A: Physical, 2026

https://www.scopus.com/pages/publications/105030493643?origin=scopusAI

33. Biomimetic Self-Sensing Hydrogel Soft Robotics: Precise Structures for Integrated Actuation and Sensing Li, S., Ma, J., Han, J., Fu, J. Advanced Functional Materials, 2026 https://www.scopus.com/pages/publications/105032567217?origin=scopusAI

34. Optimization of Central Pattern Generator-Based Locomotion Controller for Fish Robot Using Deep Deterministic Policy Gradient Vu, Q.T., Pham, M.H., Nguyen, V.D., (...), Nguyen, T.T. Lecture Notes in Networks and Systems, 2023 https://www.scopus.com/pages/publications/85145097339?origin=scopusAI

35. A reinforcement learning-based control approach with lightweight feature for robotic fish heading control in complex environments: Real-world training Chen, X., Tian, X., Chen, J., (...), Tang, S. Ocean Engineering, 2025 https://www.scopus.com/pages/publications/105007028236?origin=scopusAI

36. An Improved Backstepping Controller with an LESO and TDs for Robust Underwater 3D Trajectory Tracking of a Turtle-Inspired Amphibious Spherical Robot Hou, X., Li, Z., Guo, S., (...), Yin, H. Machines, 2022 https://www.scopus.com/pages/publications/85132013803?origin=scopusAI

37. An Efficient MPC-CPG Bionic Controller for Periodic Motions and Bounded Transitions Li, X., Wang, S., Liu, Y., (...), Zhang, Z. International Journal of Control, Automation and Systems, 2024 https://www.scopus.com/pages/publications/85191335266?origin=scopusAI

38. Trajectory tracking and obstacle avoidance control for a bionic robotic fish based on NDOB and online parameter adaptation Wang, X., Wang, M., Liu, X., (...), Yu, J. Ocean Engineering, 2026 https://www.scopus.com/pages/publications/105037468934?origin=scopusAI

39. Hydrodynamic Characteristics of Flexible Lateral Fins of Bio-robotic Fish Yang, Z., Qiu, H., Chen, Z., (...), Ren, S. Shuidonglixue Yanjiu yu Jinzhan/Chinese Journal of Hydrodynamics Ser. A, 2026 https://www.scopus.com/pages/publications/105046359336?origin=scopusAI

40. Computational Model Construction and Analysis of the Hydrodynamics of a Rhinoptera Javanica Huang, Q., Zhang, D., Pan, G. IEEE Access, 2020 https://www.scopus.com/pages/publications/85079789046?origin=scopusAI

41. The effects of undulating patterns of the dorsal fin on seahorse swimming performance Chen, G., Li, X. Physics of Fluids, 2024 https://www.scopus.com/pages/publications/85213431013?origin=scopusAI

42. Speed and Energy Efficiency of a Fish Robot Featuring Exponential Patterns of Control Tanev, I. Actuators, 2025 https://www.scopus.com/pages/publications/105001381790?origin=scopusAI

43. ROS2-Powered Autonomous Navigation for TurtleBot3: Integrating Nav2 Stack in Gazebo, RViz and Real-World Environments Athul Krishna, M.J., Babu, A.V., Damodaran, S., (...), Warrier, T.S. Proceedings - 2024 IEEE International Conference on Signal Processing, Informatics, Communication and Energy Systems: Harmonizing Signals, Data, and Energy: Bridging the Digital Future, SPICES 2024, 2024 https://www.scopus.com/pages/publications/85215277798?origin=scopusAI

44. A Crayfish-Inspired Sensor Fusion Platform for Super Additive Integration of Visual, Chemical, and Tactile Information Sakib, N.U., Karim Sadaf, M.U., Pannone, A., (...), Das, S. Nano Letters, 2024 https://www.scopus.com/pages/publications/85194931533?origin=scopusAI

45. Dynamic Load Regulation of Robots with Multi-sensor Fusion Cao, X., Tan, B., Li, Y., Ding, S. Journal of Physics: Conference Series, 2022 https://www.scopus.com/pages/publications/85145239900?origin=scopusAI

46. Electro-SLAM: Distributed underwater multi-robot SLAM via bio-inspired active and passive electro-sensing Yang, Q., Zheng, J., Wang, C., (...), Xie, G. International Journal of Robotics Research, 2026 https://www.scopus.com/pages/publications/105027855695?origin=scopusAI

47. VID-SLAM: A Robust Visual-Inertial-DVL Tightly Coupled Localization Method for Underwater Robots Wu, C., Song, Z., Li, Q., (...), Luo, J. IEEE Internet of Things Journal, 2026 https://www.scopus.com/pages/publications/105034501198?origin=scopusAI

48. PL-VAP: A Tightly Coupled Self-Localization Framework for Underwater Robots Using Point-Line Features and Visual-Acoustic-Pressure Sensor Fusion Ou, Y., Xia, S., Fan, J., (...), Hou, Z. IEEE/ASME Transactions on Mechatronics, 2025 https://www.scopus.com/pages/publications/105020739805?origin=scopusAI

49. Tightly Coupled Monocular-Inertial-Pressure Sensor Fusion for Underwater Localization of a Biomimetic Robotic Manta Ma, S., Wang, J., Huang, Y., (...), Wu, Z. IEEE Transactions on Instrumentation and Measurement, 2024 https://www.scopus.com/pages/publications/85197049737?origin=scopusAI

50. Learning Agile Swimming: An End-to-End Approach Without CPGs Lin, X., Liu, X., Wang, Y. IEEE Robotics and Automation Letters, 2025 https://www.scopus.com/pages/publications/85215288730?origin=scopusAI

51. Autonomous Vision-Based Navigation and Stability Augmentation Control of a Biomimetic Robotic Hammerhead Shark Yan, S., Wang, J., Wu, Z., (...), Yu, J. IEEE Transactions on Automation Science and Engineering, 2024 https://www.scopus.com/pages/publications/85161061267?origin=scopusAI

52. Toward the Intelligent, Safe Exploration of a Biomimetic Underwater Robot: Modeling, Planning, and Control Wang, Y., Wang, J., Yu, L., (...), Yu, J. Biomimetics, 2024 https://www.scopus.com/pages/publications/85188684373?origin=scopusAI

53. Optimizing Underwater Robot Navigation: A Study of DRL Algorithms and Multi-Modal Sensor Fusion Deowan, M.E., Yeasher Yousha, M.S., Hossain, T.M., (...), Marxer, R. Proceedings - IEEE International Conference on Robotics and Automation, 2025 https://www.scopus.com/pages/publications/105016576180?origin=scopusAI

54. Adapted D[∗]Lite to Improve Guidance, Navigation and Control of a Tail-Actuated Underwater Vehicle in Unknown Environments Algarin-Pinto, J.A., Garza-Castanon, L.E., Vargas-Martinez, A., Minchala-Avila, L.I. IEEE Access, 2023 https://www.scopus.com/pages/publications/85163510690?origin=scopusAI

55. Physics-Informed Neural Network for Flow Simulation Around a Circular Cylinder Clemons, S., Smith, S.T. Regional Student Conferences, 2026, 2026 https://www.scopus.com/pages/publications/105044013795?origin=scopusAI

56. Enhancing the accuracy of physics-informed neural networks for indoor airflow simulation with experimental data and Reynolds-averaged Navier-Stokes turbulence model Zhang, C., Wen, C.-Y., Jia, Y., (...), Li, Z. Physics of Fluids, 2024 https://www.scopus.com/pages/publications/85197386557?origin=scopusAI

57. Physics-informed neural networks modelling for systems with moving immersed boundaries: Application to an unsteady flow past a plunging foil Sundar, R., Majumdar, D., Lucor, D., Sarkar, S. Journal of Fluids and Structures, 2024 https://www.scopus.com/pages/publications/85182510190?origin=scopusAI

58. AI-empowered computational fluid dynamics in ship and ocean engineering: A review of data, algorithms and validation Ni, M., Liu, J., Zeng, Q., (...), Cai, W. Ocean Engineering, 2026 https://www.scopus.com/pages/publications/105043227220?origin=scopusAI

59. Fast Aquatic Swimmer Optimization with Differentiable Projective Dynamics and Neural Network Hydrodynamic Models Nava, E., Zhang, J.Z., Michelis, M.Y., (...), Katzschmann, R.K. Proceedings of Machine Learning Research, 2022 https://www.scopus.com/pages/publications/85146850111?origin=scopusAI

60. Data-driven reduced-order modeling for undulating swimmers Saeed, A., Saeed, H., Li, J. Physics of Fluids, 2025 https://www.scopus.com/pages/publications/105010067677?origin=scopusAI

61. Multi-Viscosity Physics-Informed Neural Networks for Generating Ultra High Resolution Flow Field Data Zhang, S., Guo, X.-W., Li, C., (...), Zhong, Y. International Journal of Computational Fluid Dynamics, 2023 https://www.scopus.com/pages/publications/85180506092?origin=scopusAI

62. Benchmarking Quantum-Assisted PINN (QA-PINN) for Computational Fluid Dynamics Shah, J., Lineswala, R., Chopra, A. Proceedings - IEEE Quantum Week 2024, QCE 2024, 2024 https://www.scopus.com/pages/publications/85217427645?origin=scopusAI

63. Stagewise edge expansion physics-informed neural networks for solving PDEs Zhang, S., Yuan, P., Pan, M., Zhang, C. Neurocomputing, 2025 https://www.scopus.com/pages/publications/105013576772?origin=scopusAI

64. Evolution of physics-informed neural networks: Recent architectural variants and optimization strategies Ahmad, Zafar, H., Zafar, A., (...), Ahmed, K.K. Array, 2026 https://www.scopus.com/pages/publications/105028242826?origin=scopusAI

65. Modelling of Underwater Vehicles using Physics-Informed Neural Networks with Control Amer, A., Felsager, D., Brodskiy, Y., Sarabakha, A. Proceedings of the International Joint Conference on Neural Networks, 2025 https://www.scopus.com/pages/publications/105023986129?origin=scopusAI

66. Research on model predictive control of autonomous underwater vehicle based on physics informed neural network modeling Liu, T., Zhao, J., Huang, J., (...), Zhao, B. Ocean Engineering, 2024 https://www.scopus.com/pages/publications/85190471534?origin=scopusAI

67. Domain-Decoupled Physics-informed Neural Networks with Closed-Form Gradients for Fast Model Learning of Dynamical Systems Krauss, H., Habich, T.-L., Seel, T., (...), Schappler, M. Proceedings of the International Conference on Informatics in Control, Automation and Robotics, 2024 https://www.scopus.com/pages/publications/105001299316?origin=scopusAI

68. TO-SoFiT: Topology Optimization of Hydraulic Soft Fish Tail Design for programmable undulating locomotion Padmaprabhan, A., Shaji, A., Kumar, P. AIR 2025 - 7th International Conference of the Robotics Society India, 2026 https://www.scopus.com/pages/publications/105045296344?origin=scopusAI

69. Bioinspired hydrogel jellyfish with mechanical flexibility and acoustic transparency Zhang, J., Zhang, T., Dong, E., (...), Zhang, Y. Cell Reports Physical Science, 2022 https://www.scopus.com/pages/publications/85140227253?origin=scopusAI

70. An Invisible Bionic Dragonfly Based on Fully-Transparent Conductive Hydrogel and Dielectric Elastomer He, R., Tao, K., Chen, Z., (...), Chang, H. 21st International Conference on Solid-State Sensors, Actuators and Microsystems, TRANSDUCERS 2021, 2021 https://www.scopus.com/pages/publications/85114964068?origin=scopusAI

71. Improving Efficiency and Maneuverability through Design of a Robotic Fish with Independent Control of Tail and Caudal Fin Deflection Walker, K.L., Obayashi, N., Marques, G.V., Hughes, J. ROBOSOFT 2026 - 9th IEEE RAS International Conference on Soft Robotics, 2026 https://www.scopus.com/pages/publications/105041904542?origin=scopusAI

72. Design, Modeling, and Optimization of Hydraulically Powered Double-Joint Soft Robotic Fish Liu, S., Liu, C., Wei, G., (...), Ren, L. IEEE Transactions on Robotics, 2025 https://www.scopus.com/pages/publications/85214559701?origin=scopusAI

73. 4D printing of pneumatic soft robots sensors and actuators Long, F., Zhang, Z., Wang, B., (...), Xu, G. Smart Materials in Additive Manufacturing: 4D-Printed Robotic Materials, Sensors, and Actuators, Volume 3, 2024 https://www.scopus.com/pages/publications/85202796421?origin=scopusAI

74. A Review of Soft Robotic Actuators and Their Applications in Bioengineering, with an Emphasis on HASEL Actuators' Future Potential Perera, O., Liyanapathirana, R., Gargiulo, G., Gunawardana, U. Actuators, 2024 https://www.scopus.com/pages/publications/85213316627?origin=scopusAI

75. Water hydraulic soft actuators for underwater autonomous robotic systems Chen, G., Yang, X., Zhang, X., Hu, H. Applied Ocean Research, 2021 https://www.scopus.com/pages/publications/85100373080?origin=scopusAI

76. Hydraulic variable stiffness mechanism for swimming locomotion optimization of soft robotic fish Ju, I., Yun, D. Ocean Engineering, 2023 https://www.scopus.com/pages/publications/85167794536?origin=scopusAI

77. Dynamics Modeling and Oscillation Control of a Biomimetic Fish-tail Robot Based on a Three-element Model Meng, Q.-X., Xiong, K., Yan, Z., (...), Su, C.-Y. Zidonghua Xuebao/Acta Automatica Sinica, 2026 https://www.scopus.com/pages/publications/105044409824?origin=scopusAI

78. A versatile control architecture based on reinforcement learning for diverse control tasks in robotic fish Chen, X., Tian, X., Liu, X., (...), Cao, Q. Ships and Offshore Structures, 2025 https://www.scopus.com/pages/publications/105019658448?origin=scopusAI

79. Acoustics-Driven Performance Enhancement in Underwater Vehicles: From Component Innovation to Intelligent Actuation Wang, X., Wang, Z., Chen, L., (...), Chen, C. Actuators, 2026 https://www.scopus.com/pages/publications/105037212422?origin=scopusAI

80. Research and Application Progress of Distributed Fiber Optic Hydrophone Technology Zhaoyong, W., Yifan, L., Yici, C., (...), Haiwen, C. Guangxue Xuebao/Acta Optica Sinica, 2024 https://www.scopus.com/pages/publications/85182734499?origin=scopusAI

81. Acoustic Wake-Up Technology for Microsystems: A Review Yang, D., Zhao, J. Micromachines, 2023 https://www.scopus.com/pages/publications/85146593236?origin=scopusAI

82. Bull shark-inspired robotic fish enabled by synergistic variable-stiffness mechanism, fluidic propulsion, and autonomous control Ding, J., Ye, X., Zhang, L., (...), Ding, J. Sensors and Actuators A: Physical, 2026 https://www.scopus.com/pages/publications/105048083292?origin=scopusAI

83. Study on the propulsion performance of robotic fish driven by pectoral fins in burst-and-coast swimming Li, B., Li, Z., Wang, Z., (...), Nie, Y. Journal of Ocean Engineering and Science, 2026 https://www.scopus.com/pages/publications/105034878194?origin=scopusAI

84. Biomimetic omnidirectional multi-tail underwater robot Li, Z., Chao, X., Hameed, I., (...), Jing, X. Mechanical Systems and Signal Processing, 2022 https://www.scopus.com/pages/publications/85127218595?origin=scopusAI

85. Numerical Investigation of Dimensionless Parameters in Carangiform Fish Swimming Hydrodynamics Macías, M.M., García-Ortiz, J.H., Oliveira, T.F., Brasil Junior, A.C.P. Biomimetics, 2024 https://www.scopus.com/pages/publications/85183144498?origin=scopusAI

86. Review of Computational Fluid Dynamics Analysis in Biomimetic Applications for Underwater Vehicles Zhang, Z., Wang, Q., Zhang, S. Biomimetics, 2024 https://www.scopus.com/pages/publications/85185947715?origin=scopusAI

87. Bio-inspired profile improving the flow near the stern of an underwater vehicle Xie, Z., Liu, Y., Wang, S., He, G. Ocean Engineering, 2024 https://www.scopus.com/pages/publications/85208386865?origin=scopusAI

88. Deep-Sea Biomimetic Manta Ray Robots: A Comprehensive Review Based on Operational Depth Spectrum, Structures, Energy Optimization, and Control Systems Ye, L., Liu, H., Ding, Q., (...), Fan, D. Biomimetics, 2026 https://www.scopus.com/pages/publications/105033895852?origin=scopusAI

89. Salmon behavioural response to robots in an aquaculture sea cage Kruusmaa, M., Gkliva, R., Tuhtan, J.A., (...), Alfredsen, J.A. Royal Society Open Science, 2020 https://www.scopus.com/pages/publications/85083201082?origin=scopusAI

90. Edible aquatic robots with Marangoni propulsion Zhang, S., Kwak, B., Zhu, R., (...), Floreano, D. Nature Communications , 2025 https://www.scopus.com/pages/publications/105004458956?origin=scopusAI

91. The Use of Robots in Aquatic Biomonitoring with Special Focus on Biohybrid Entities Rajewicz, W., Schmickl, T., Thenius, R. Mechanisms and Machine Science, 2022 https://www.scopus.com/pages/publications/85129256126?origin=scopusAI

92. Application of Artificial Intelligence in Biomimetic Underwater Vehicles – A Review Yu, S., Wang, Y., Zhang, H., (...), Li, F. Journal of Bionic Engineering, 2026 https://www.scopus.com/pages/publications/105035188009?origin=scopusAI

93. Research Review on Underwater Robotics in Fisheries Zhou, H., Zhu, R., Liu, J. Nongye Jixie Xuebao/Transactions of the Chinese Society for Agricultural Machinery, 2025 https://www.scopus.com/pages/publications/105020585933?origin=scopusAI

94. Deep-Sea Soft Bionic Fish: Advances in Pressure-Tolerant Design, Soft Actuation, and Autonomous Systems Yang, S., Liu, H., Tang, D. Biomimetics, 2026 https://www.scopus.com/pages/publications/105045707453?origin=scopusAI

95. Pectoral Fin-Assisted Braking and Agile Turning: A Biomimetic Approach to Improve Underwater Robot Maneuverability He, Q., Zhu, Y., Li, W., (...), Fan, D. Journal of Marine Science and Engineering, 2025 https://www.scopus.com/pages/publications/105011631512?origin=scopusAI

96. Soft Materials and Devices Enabling Sensorimotor Functions in Soft Robots Su, J., He, K., Li, Y., (...), Chen, X. Chemical Reviews, 2025 https://www.scopus.com/pages/publications/105001494916?origin=scopusAI

97. A review and case study on Underwater Legged Robot simulation Aguirregomezcorta Aina, J., Aguzzi, J., Picardi, G. Ocean Engineering, 2026 https://www.scopus.com/pages/publications/105045681553?origin=scopusAI

98. From Remote Sensing to Artificial Intelligence in Coral Reef Monitoring Piñeros, V.J., Reveles-Espinoza, A.M., Monroy, J.A. Machines, 2024 https://www.scopus.com/pages/publications/85207675064?origin=scopusAI